

What are CSS Selectors?

-> The CSS selectors are used to target(select) the HTML elements.

-> We apply CSS styles using selectors.

Types of CSS Selectors:

-> There are 5 main types:

1. Simple Selectors
2. Combinators Selectors
3. Attribute Selectors
4. Pseudo-Class Selectors
5. Pseudo-Element Selectors

1. Simple Selectors:

-> Simple selectors are divided into main 4 types:

1. Id Selector (#) → Selects a unique element using its id.
2. Class Selector (.) → Selects multiple elements with same class.
3. Tagname or Element name Selector (tagName) -> → Selects elements by tag name (like p, h1).
4. Universal Selector (*) → Selects all elements on the page.

Priority Order for Simple Selectors: Id > Class > Tagname > Universal

2. Combinators Selectors:

-> The combination of more than one selector is called combinators selectors.

-> There are 5 types of combinators selectors are as follows:

1. Descendant Selector (space) → Selects elements inside another element (any level).
2. Child Selector (>) → Selects direct child elements only.
3. Adjacent Sibling Selector (+) → Selects the immediate next element.
4. General Sibling Selector (~) → Selects all next sibling elements.
5. Grouping Selector (,) → Used to apply the same CSS style to multiple elements at once.

3. Attribute Selector:

-> Attribute Selector [] → Selects elements based on their attribute or attribute value.

-> Types:

1. [attribute] → Selects elements that have the attribute
2. [attribute="value"] → Selects elements with exact value
3. [attribute~="value"] → Value contains a word
4. [attribute|="value"] → Value is exact or starts with value-
5. [attribute^="value"] → Value starts with given value
6. [attribute\$="value"] → Value ends with given value
7. [attribute*="value"] → Value contains given value

4. Pseudo-Class Selectors:

-> Pseudo-Class Selector (:) → Selects elements based on their state (like hover, focus, click).

-> Types:

1. Dynamic
 - 1.1 :hover → when mouse is over element
 - 1.2 :active → when element is clicked

1.3 :focus → when input is selected

1.4 :visited → visited link

1.5 :link → unvisited link

2. Structural

2.1 :first-child → first child element

2.2 :last-child → last child element

2.3 :nth-child(n) → specific position

2.4 :first-of-type → first of that type

2.5 :last-of-type → last of that type

3. UI state

3.1 :enabled → enabled element

3.2 :disabled → disabled element

3.3 :checked → selected checkbox/radio

5. Pseudo-Element Selectors:

-> Pseudo-Element Selector (::) → Selects and styles a specific part of an element.

-> Types:

1. ::first-line → first line of text

2. ::first-letter → first letter

3. ::before → content before element

4. ::after → content after element

5. ::marker → list item marker

6. ::selection → selected text

7. ::placeholder → placeholder text inside input/textarea